PIRCHE®

Connect to the PIRCHE API with R

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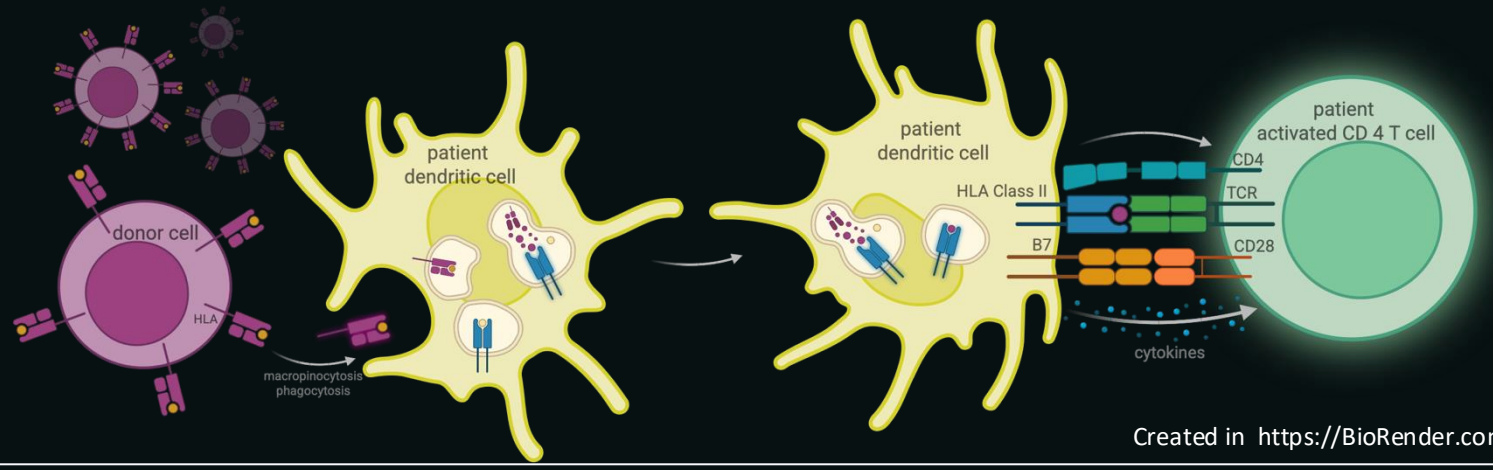
PIRCHE-T2 (previously PIRCHE-II)

Indirect recognition of digested HLA

SOT PIRCHE-T2 score is the number of distinct donor HLA derived peptides that are predicted to bind to recipient HLA Class II molecules, not present in the recipient peptidome

CD4+ T helper cells

- Stimulate CD8+ cytotoxic T cells
- Promote B cell antibody production



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PIRCHE-B (previously Snow)

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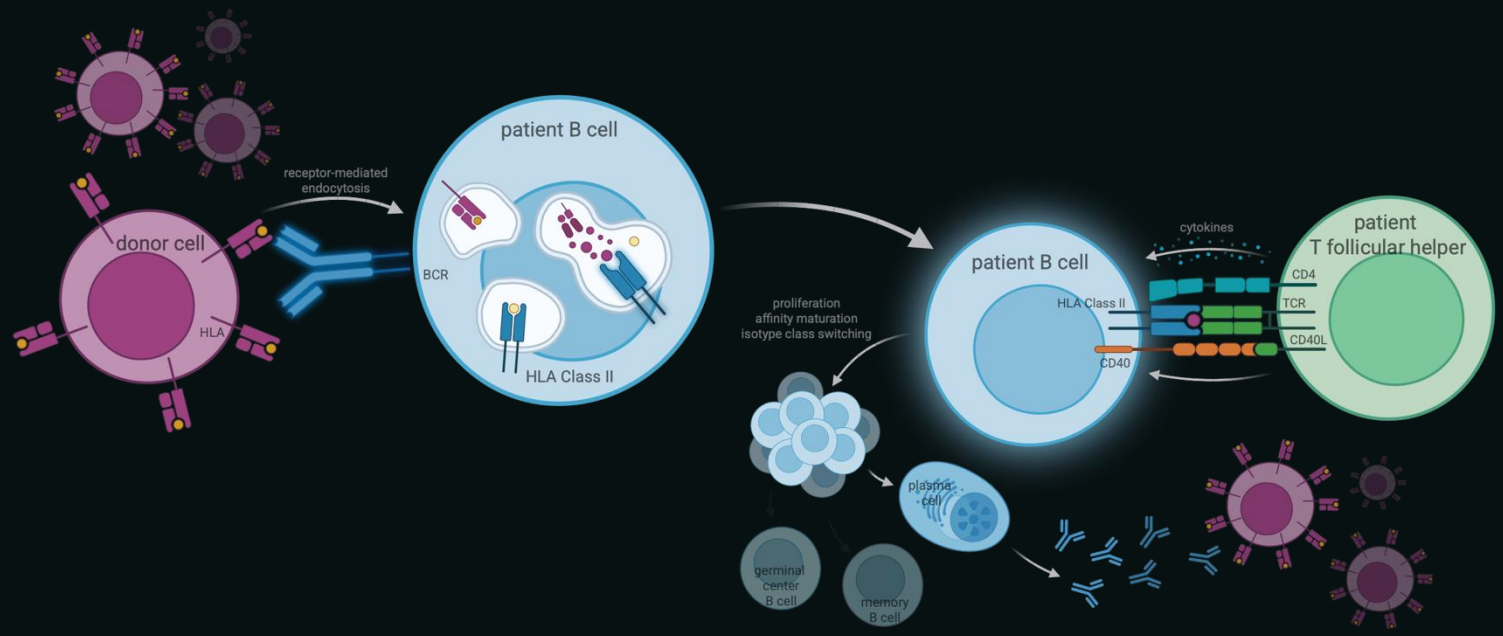
Direct B cell recognition of a region on intact HLA protein

SOT PIRCHE-B score is the number of interlocus donor-specific amino acid mismatches that are surface accessible.

Snow:

- Combination of two algorithms predict protein-specific surface area (Snowflake) and protein-specific local protrusion rank (Snowball).
- Allele-specific
  - Neural network predictions applied to each HLA protein, therefore accounting for allele-specific changes to protein structure



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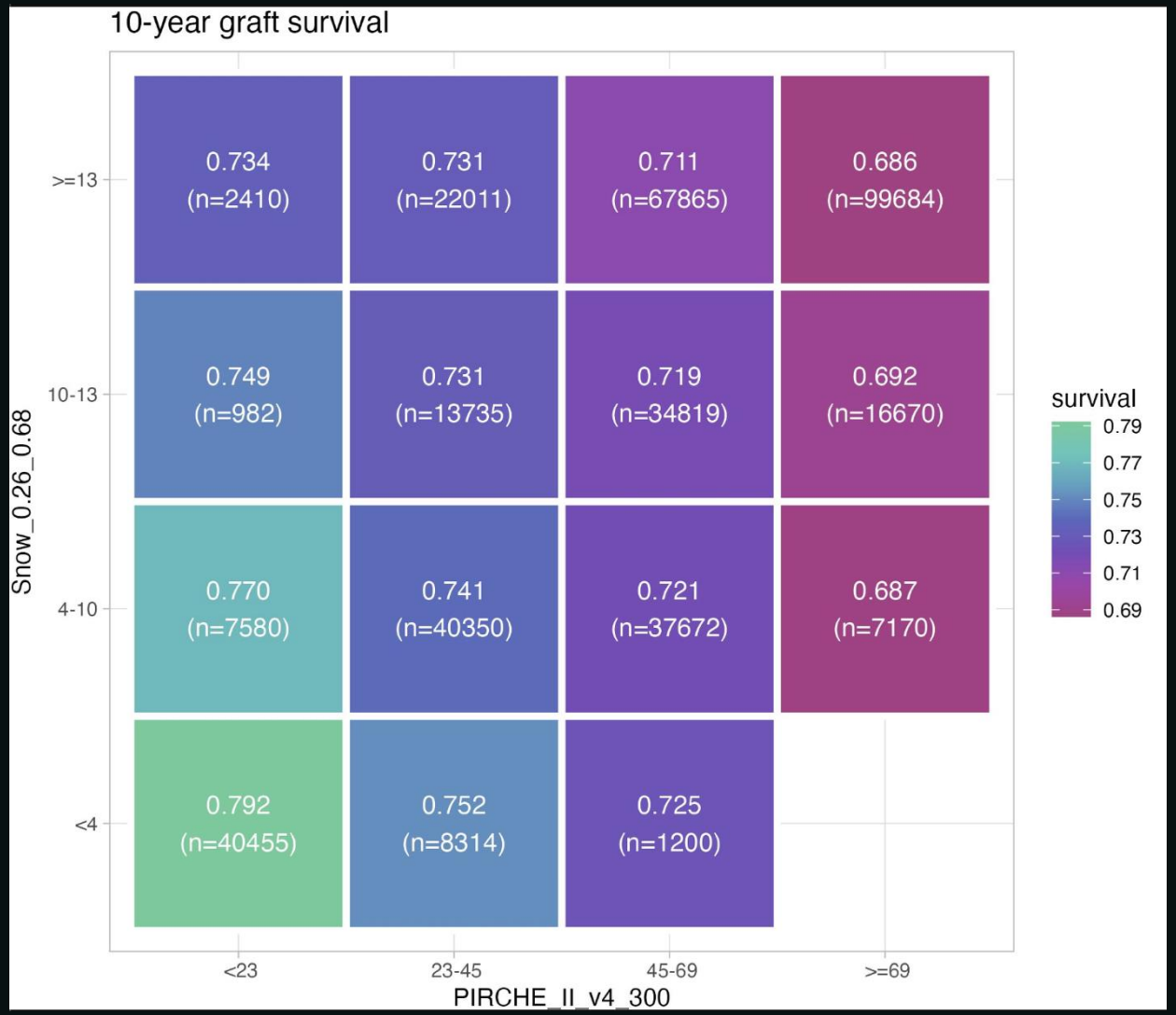
Clinical Correlation



Niemann M, et al. Advancing risk stratification in kidney transplantation: integrating HLA-derived T-cell epitope and B-cell epitope matching algorithms... Front. Immunol. (2025).

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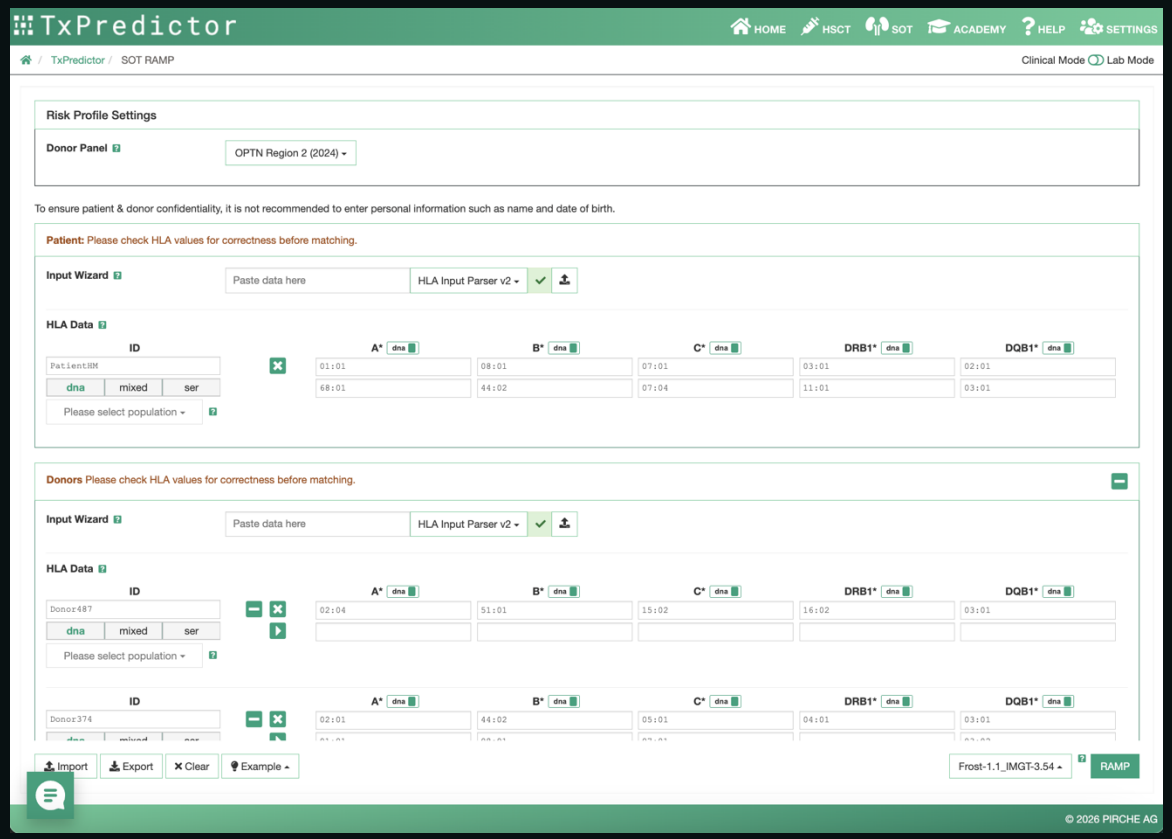
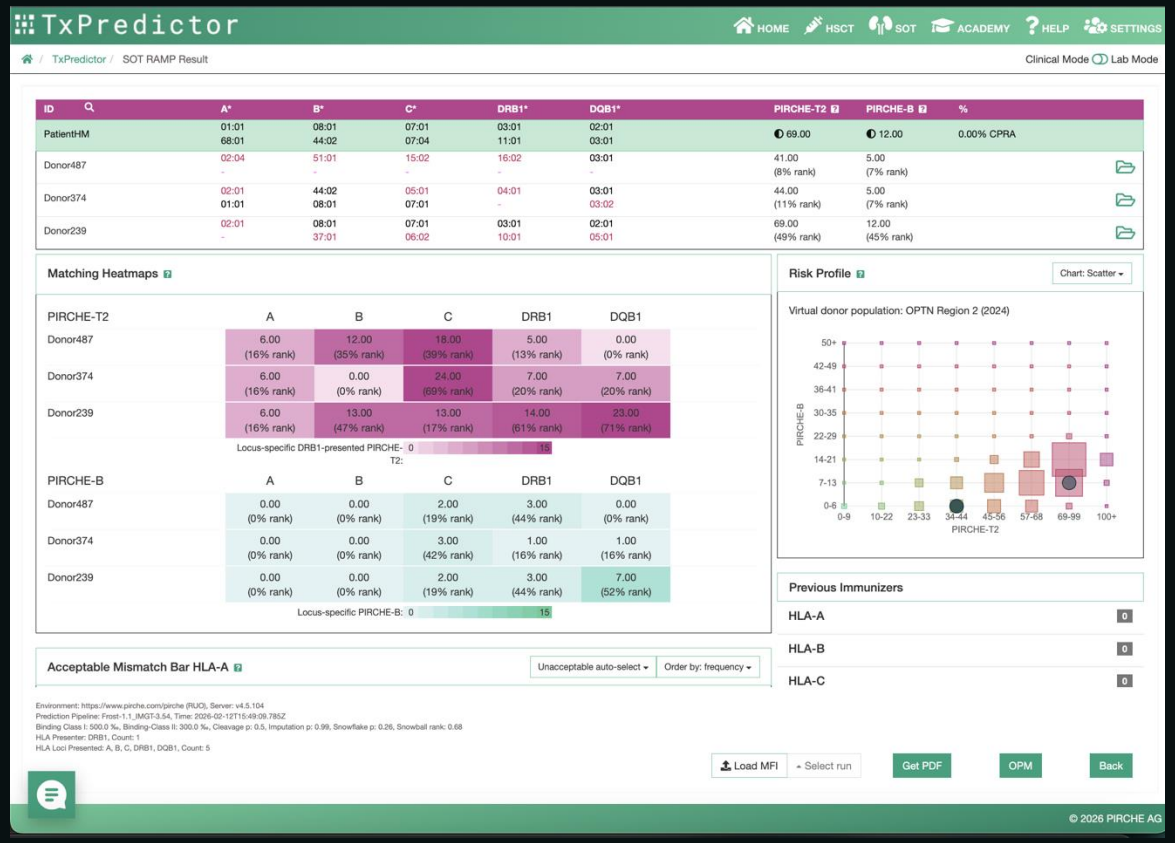
TxPredictor – Web Application

TxPredictor contains several matching modules designed for various transplant scenarios

- Calculate PIRCHE-T2 and PIRCHE-B scores using just the HLA typings.
- Advanced features allow for a “deep dive” to personalize patient care.





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Automated Workflow

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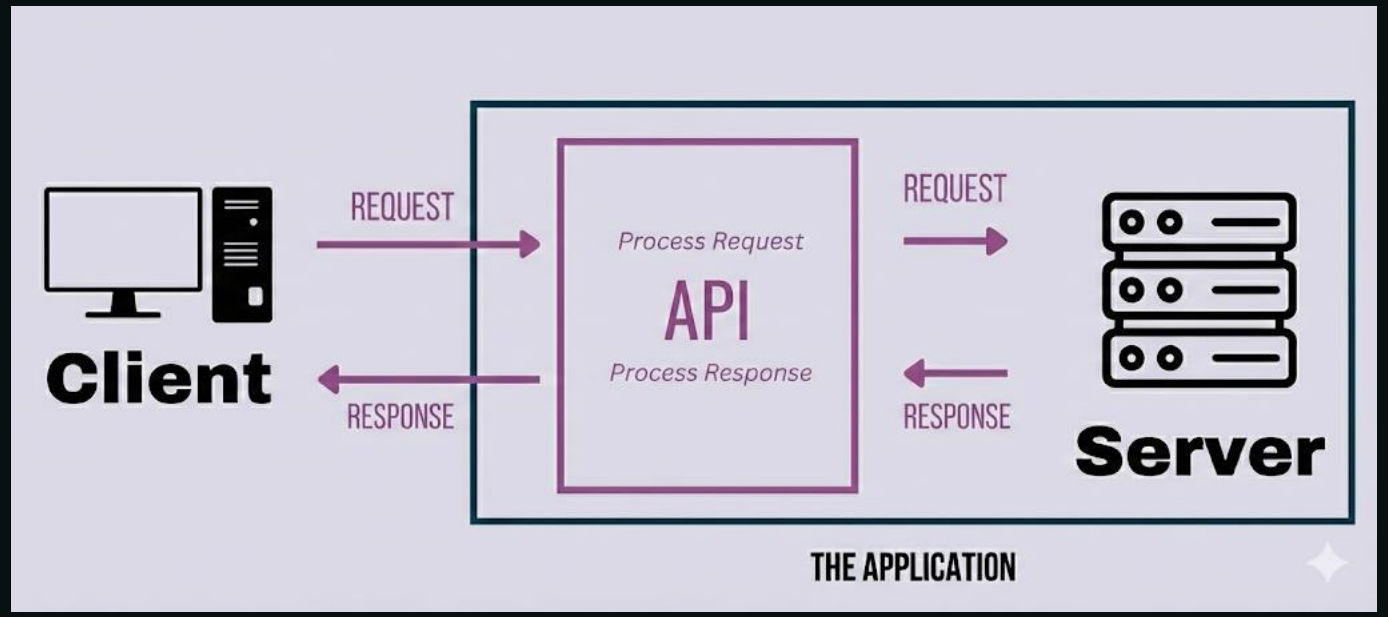
API (Application Programming Interface)

allows a client to communicate with a server.



Implementation Context

- Client – Laboratory Software
  - (e.g. custom VXM, partner integration, today's R script)
- Server – PIRCHE Web Service

Figure adapted from Codefinity

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Structure of an API REST Request

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Endpoint – defines a protocol using two main parts:

- URL: specific URL path
  - Base URL: https://research.pirche.com/pirche/rest
    - PIRCHE-T2: /sot/api/match
    - PIRCHE-B: /sot/snow/api/match
- HTTP Method: **POST**

Requirements – information required by the specified 'protocol':

- Header:
  - Authorization: 'login' code
  - Content-Type: language/format (e.g. application/json)
- Payload:
  - Input data: patient/donor IDs and HLA typings
  - Parameters: prediction pipeline

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JSON

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Each ‘protocol’ expects and returns specific information in a structured format

JSON (JavaScript Object Notation):

- A text format that uses brackets to organize data into a hierarchy (like a family tree).

Key Syntax:

- { } Curly Braces: Represent a Single Object (e.g., one patient).
- [ ] Square Brackets: Represent a List (e.g., list of donors).
- "Key": "Value": Defines the data
  - Key: Label (used to extract data in R).
  - Value: Data Value (Number, String, or List).

```
curl -X POST "https://research.pirche.com/pirche/rest/sot/snow/api/match" \
  -H "Authorization: bearer cd42f0b9-a9f2-4dc4-b0da-ac6aad990320" \
  -H "Content-Type: application/json;charset=utf-8" \
  -H "Accept: application/json" \
  -d '{
    "databaseVersion": "Frost-1.1_IMGT-3.54",
    "hlaValueFormatVersion": "v2",
    "patient": {
      "id": "IHW09007",
      "population": "NMDP EUR haplotypes (2007)",
      "glString": "A*02^C*06^B*57^DRB1*04:01+DRB1*16:02^DQB1*03:02+DQB1*05"
    },
    "donors": [
      {
        "id": "IHW09011",
        "population": "NMDP EUR haplotypes (2007)",
        "glString": "A*01^C*12:02^B*52^DRB1*15^DQB1*06"
      },
      {
        "id": "IHW09012",
        "population": "NMDP EUR haplotypes (2007)",
        "glString": "A*02^C*07^B*57^DRB1*16^DQB1*05"
      }
    ]
  }'
```

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R Script Workflow

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R handles the JSON structure easily, while allowing us to create a fully customized workflow that retrieves PIRCHE scores in just one step.

Authorization:

- Send API key
- Receive a temporary Bearer Token to be used in the POST request

POST Request:

- Send the Header and Payload to the Endpoint
  - Header contains the Method (POST) and Bearer Token
  - Payload contains Patient & Donor data
    - HLA typings are formatted into GL Strings using immunogenetr

Response:

- Server calculates PIRCHE scores using the information sent in the Payload
- Results are returned in a JSON object
- R is used to parse scores and format into a readable table

GitHub https://github.com/PIRCHE/pipeline.git

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API Examples in R Notebook

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